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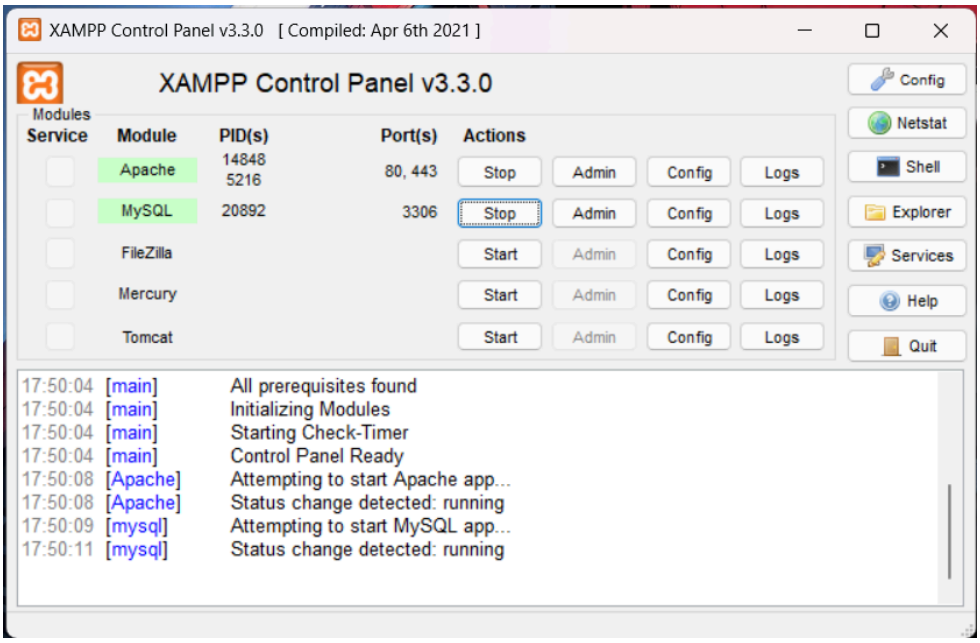
NIM : 242410101079
Kelas : A

LAPORAN PEMROGRAMAN WEBSITE

Aktivitas Mandiri 1

1. Instal XAMPP terbaru, pastikan PHP 8.2+ tersedia (cek: php --version di terminal)




```
C:\Users\CHELSEA>php --version
PHP 8.3.30 (cli) (built: Jan 13 2026 22:50:40) (ZTS Visual C++ 2019 x64)
Copyright (c) The PHP Group
Zend Engine v4.3.30, Copyright (c) Zend Technologies
```

Xampp sudah di install dan sudah dilakukan pengecekan versi php yaitu 8.3

2. Instal Composer dari getcomposer.org, verifikasi dengan: composer --version

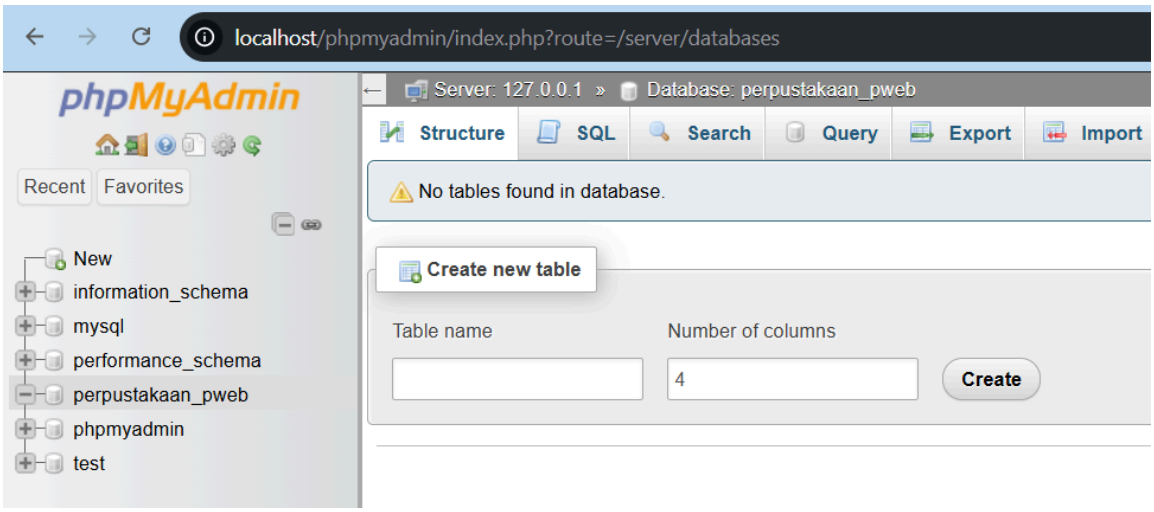
```
C:\Users\CHELSEA>composer --version
Composer version 2.9.4 2026-01-22 14:08:50
PHP version 8.3.30 (C:\laragon\bin\php\php-8.3.30-Win32-vs16-x64\php.exe)
Run the "diagnose" command to get more detailed diagnostics output.
```

3. Buat proyek Laravel baru: composer create-project laravel/laravel siak-app

```
C:\xampp\htdocs>composer create-project laravel/laravel perpustakaan-pweb
Creating a "laravel/laravel" project at "./perpustakaan-pweb"
Installing laravel/laravel (v13.3.0)
  - Installing laravel/laravel (v13.3.0): Extracting archive
Created project in C:\xampp\htdocs\perpustakaan-pweb
> @php -r "file_exists('.env') || copy('.env.example', '.env');"
Loading composer repositories with package information
Updating dependencies
```

```
60 package suggestions were added by new dependencies, use `composer suggest` to see details.
Generating optimized autoload files
> Illuminate\Foundation\ComposerScripts::postAutoloadDump
> @php artisan package:discover --ansi

  INFO  Discovering packages.

  laravel/pail ..................................................... DONE
  laravel/tinker ................................................... DONE
  nesbot/carbon .................................................... DONE
  nunomaduro/collision ............................................. DONE
  nunomaduro/termwind .............................................. DONE

78 packages you are using are looking for funding.
Use the `composer fund` command to find out more!
> @php artisan vendor:publish --tag=laravel-assets --ansi --force

  INFO  No publishable resources for tag [laravel-assets].

No security vulnerability advisories found.
> @php artisan key:generate --ansi

  INFO  Application key set successfully.

> @php -r "file_exists('database/database.sqlite') || touch('database/database.sqlite');"
> @php artisan migrate --graceful --ansi

  INFO  Preparing database.

  Creating migration table ..................................... 6.50ms DONE

  INFO  Running migrations.

  0001_01_01_000000_create_users_table ......................... 8.86ms DONE
  0001_01_01_000001_create_cache_table ......................... 5.52ms DONE
  0001_01_01_000002_create_jobs_table .......................... 7.90ms DONE
```

4. Buat database baru 'siak_db' di phpMyAdmin



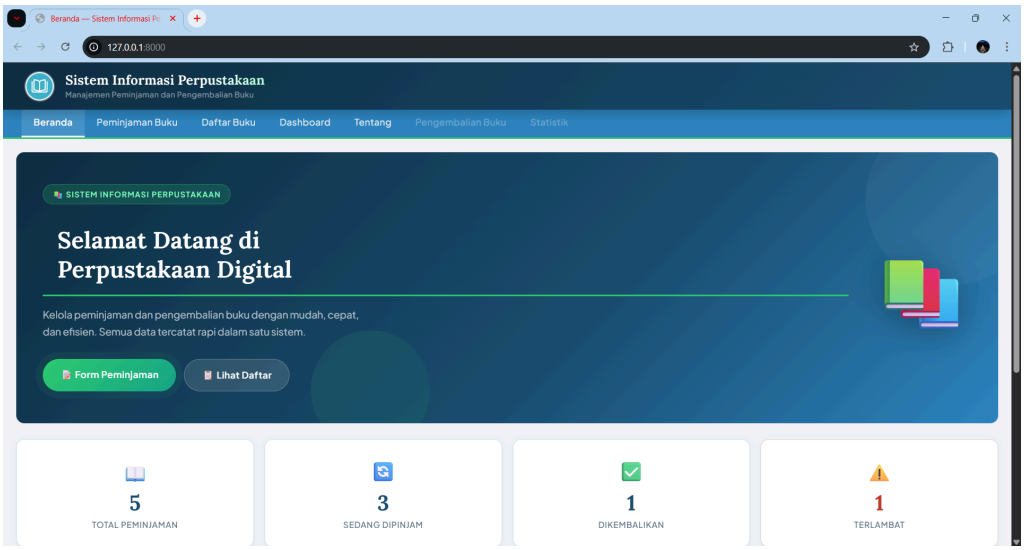
5. Konfigurasi file .env: isi DB_DATABASE, DB_USERNAME, DB_PASSWORD sesuai lokal

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=perpustakaan_pweb
DB_USERNAME=root
DB_PASSWORD=
```

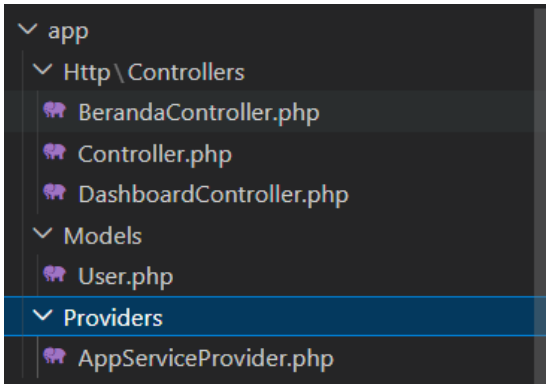
6. Jalankan: php artisan key:generate

```
PS C:\xampp\htdocs\perpustakaan-pweb> php artisan key:generate

  INFO  Application key set successfully.
```

7. Jalankan server: php artisan serve — buka http://127.0.0.1:8000 di browser

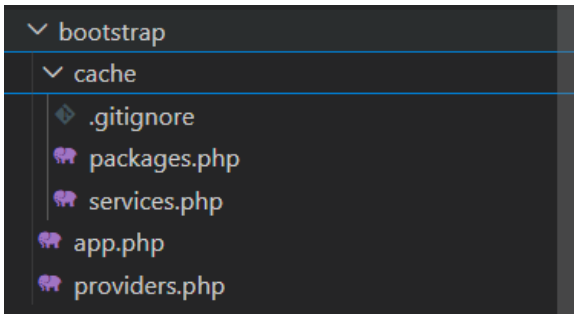


8. Eksplorasi struktur folder, buka setiap folder dan pahami fungsinya:
   - Folder App



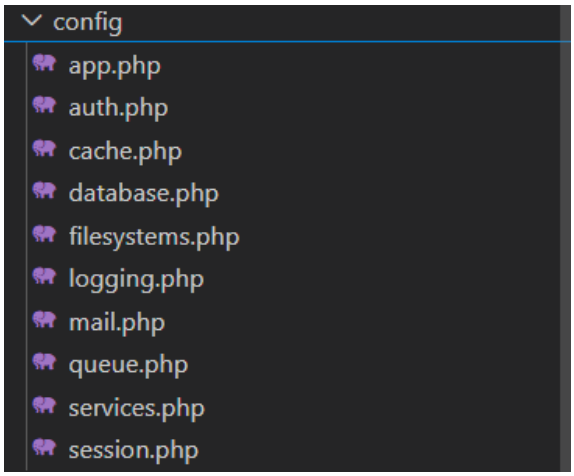
Folder ini merupakan inti dari aplikasi Laravel yang berisi logika utama seperti Controller, Model, dan Provider. Semua proses utama aplikasi biasanya dikelola di dalam folder ini.

- Folder Bootstrap



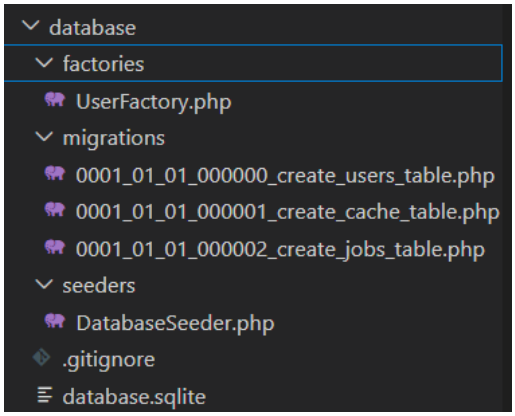
Folder ini digunakan untuk proses awal menjalankan Laravel, termasuk memuat konfigurasi dasar aplikasi dan cache agar aplikasi bisa berjalan dengan cepat.

- Folder Config



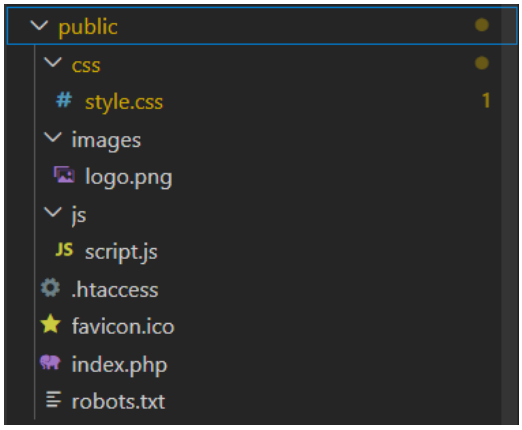
Folder ini berisi berbagai file konfigurasi aplikasi seperti pengaturan database, email, session, dan pengaturan umum lainnya.

- Folder Database



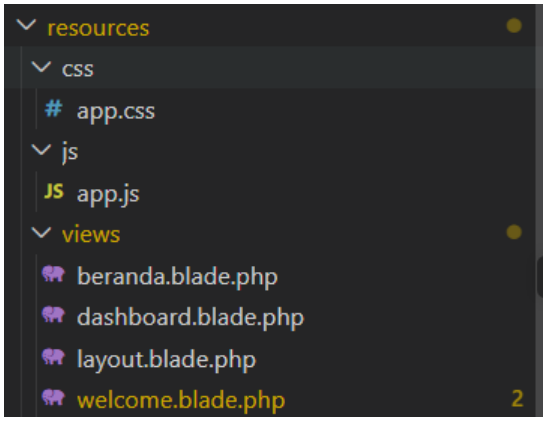
Folder ini digunakan untuk mengelola database, berisi migration (struktur tabel), seeder (data awal), dan factory (data dummy).

- Folder Public



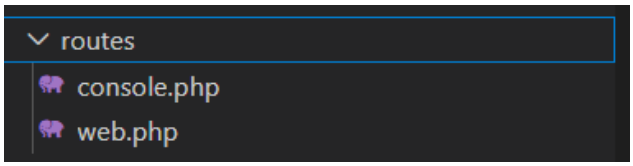
Folder ini adalah folder yang bisa diakses langsung oleh browser dan menjadi pintu masuk aplikasi melalui file index.php, serta tempat menyimpan aset seperti gambar dan file publik lainnya.

- Folder Resources



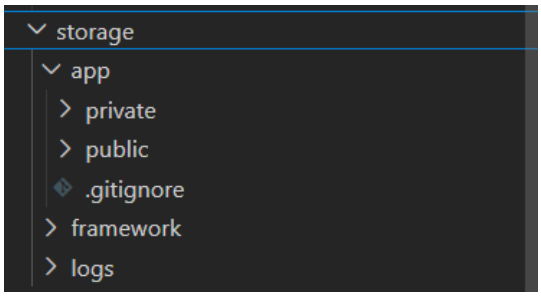
Folder ini berisi tampilan (view) aplikasi menggunakan Blade, serta file front-end seperti CSS dan JavaScript.

- Folder Routes



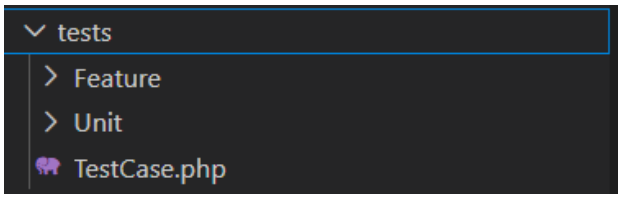
Folder ini digunakan untuk mendefinisikan rute atau URL aplikasi yang menghubungkan permintaan pengguna ke controller atau view.

- Folder Storage



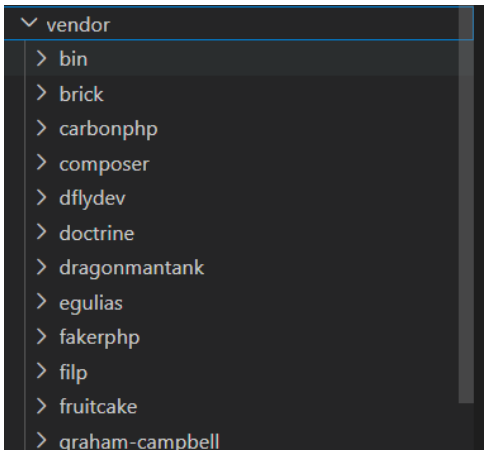
Folder ini digunakan untuk menyimpan file seperti cache, session, log, dan file upload dari pengguna.

- Folder Tests



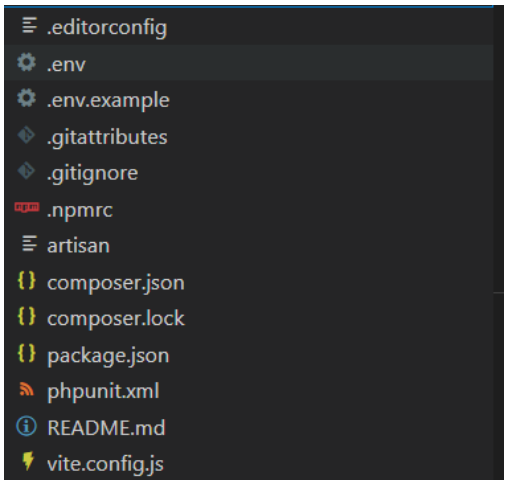
Folder ini digunakan untuk melakukan pengujian aplikasi secara otomatis untuk memastikan fitur berjalan dengan baik.

- Folder Vendor



Folder ini berisi library atau dependensi dari pihak ketiga yang diinstall melalui Composer dan digunakan oleh Laravel.

- File Lainnya (Root Project)



Selain folder utama, Laravel juga memiliki beberapa file penting di bagian root project seperti .env yang berisi konfigurasi aplikasi (database, session, dll), .env.example sebagai contoh konfigurasi, artisan untuk menjalankan perintah Laravel melalui terminal, composer.json dan composer.lock untuk mengatur dependensi PHP, package.json untuk dependensi JavaScript, .gitignore dan .gitattributes untuk pengaturan Git, phpunit.xml untuk konfigurasi testing, README.md sebagai dokumentasi proyek, serta vite.config.js untuk pengelolaan asset frontend. File-file ini berperan dalam konfigurasi, pengelolaan dependensi, dan proses pengembangan aplikasi Laravel.

9. Jalankan php artisan route:list dan catat route apa saja yang sudah terdaftar

```
PS C:\xampp\htdocs\perpustakaan-pweb> php artisan route:list

  GET|HEAD  / ..................................................................... beranda › BerandaController@index
  GET|HEAD  dashboard ........................................................... dashboard › DashboardController@index
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesystem/FilesystemServiceProv…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/Filesystem/FilesystemServ…
  GET|HEAD  up ......... vendor/laravel/framework/src/Illuminate/Foundation/Configuration/ApplicationBuilder.php:223

                                                                                                  Showing [5] routes
```

Aktivitas Mandiri 2

1. Buat Controller DashboardController: php artisan make:controller DashboardController

```
PS C:\xampp\htdocs\perpustakaan-pweb> php artisan make:controller DashboardController

  INFO  Controller [C:\xampp\htdocs\perpustakaan-pweb\app\Http\Controllers\DashboardController.php] created successfu
lly.
```

2. Tambahkan method index() yang mengembalikan view('dashboard') di DashboardController

```
PERPUSTAKAAN-PWEB
app
Http\Controllers
BerandaController.php
Controller.php
DashboardController.php
Models
Providers
bootstrap
config
database
public
resources
```

app > Http > Controllers > DashboardController.php > ...

```php
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;

class DashboardController extends Controller
{
    public function index()
    {
        return view('dashboard');
    }
}
```

3. Daftarkan route di routes/web.php: Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard')

routes > web.php

```php
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\BerandaController;
use App\Http\Controllers\DashboardController;

// Dashboard
Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');
```

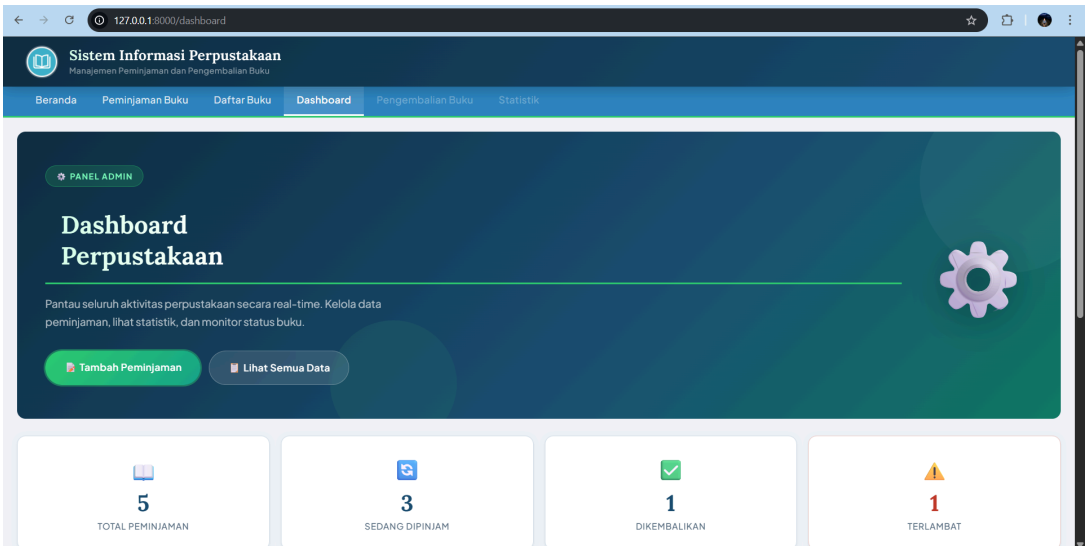
4. Buat file view sementara resources/views/dashboard.blade.php berisi '<h1>Dashboard SIAK</h1>'

resources > views > dashboard.blade.php > ...

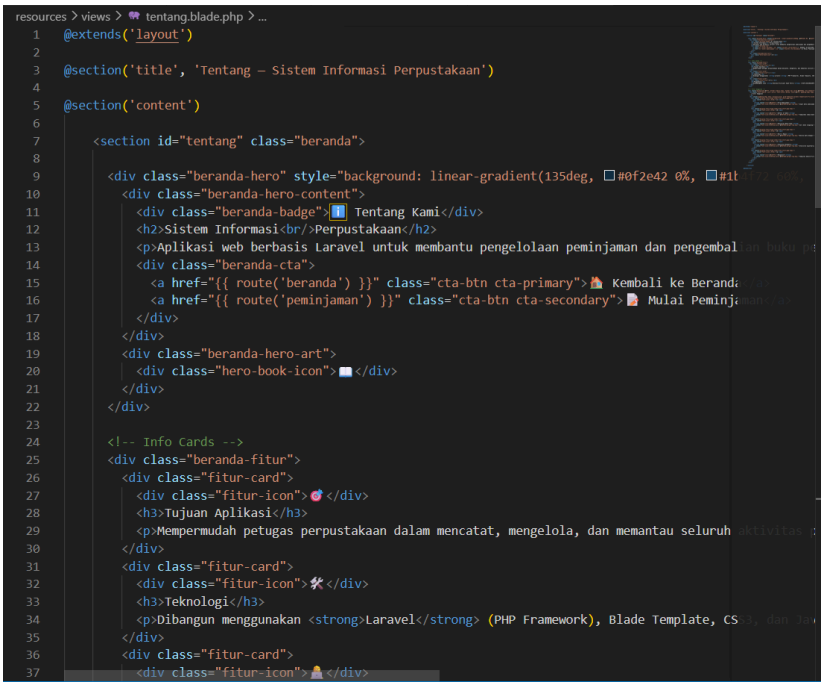
```blade
@extends('layout')

@section('title', 'Dashboard — Sistem Informasi Perpustakaan')

@section('content')

    <section id="dashboard" class="beranda">

      <div class="beranda-hero" style="background: linear-gradient(135deg, #0f2e42 0%, #1b
        <div class="beranda-hero-content">
          <div class="beranda-badge">⚙ Panel Admin</div>
          <h2>Dashboard<br/>Perpustakaan</h2>
          <p>Pantau seluruh aktivitas perpustakaan secara real-time. Kelola data peminjaman, lihat stati
          <div class="beranda-cta">
            <a href="{{ route('peminjaman') }}" class="cta-btn cta-primary">📝 Tambah Peminjaman</a>
            <a href="{{ route('daftar') }}" class="cta-btn cta-secondary">📋 Lihat Semua Data</a>
          </div>
        </div>
        <div class="beranda-hero-art">
          <div class="hero-book-icon">⚙</div>
        </div>
      </div>

      <!-- Statistik Utama -->
      <div class="beranda-stats">
        <div class="bstat-card">
          <div class="bstat-icon">📖</div>
          <div class="bstat-num" id="bstat-total">5</div>
          <div class="bstat-label">Total Peminjaman</div>
        </div>
        <div class="bstat-card">
          <div class="bstat-icon">🔄</div>
          <div class="bstat-num" id="bstat-dipinjam">2</div>
          <div class="bstat-label">Sedang Dipinjam</div>
        </div>
        <div class="bstat-card">
          <div class="bstat-icon">✅</div>
```

Disini untuk tampilan dashboard dan lain lain sudah saya sesuaikan dengan project yang saya buat yaitu Sistem Informasi Perpustakaan.

5. Akses http://127.0.0.1:8000/dashboard di browser - pastikan tampil



6. Buat route /tentang dengan Route::view('/tentang', 'tentang') dan buat view-nya



```
resources > views > tentang.blade.php > ...
@extends('layout')

@section('title', 'Tentang — Sistem Informasi Perpustakaan')

@section('content')

    <section id="tentang" class="beranda">

      <div class="beranda-hero" style="background: linear-gradient(135deg, #0f2e42 0%, #1b
        <div class="beranda-hero-content">
          <div class="beranda-badge">ℹ️ Tentang Kami</div>
          <h2>Sistem Informasi<br/>Perpustakaan</h2>
          <p>Aplikasi web berbasis Laravel untuk membantu pengelolaan peminjaman dan pengembal
          <div class="beranda-cta">
            <a href="{{ route('beranda') }}" class="cta-btn cta-primary">🏠 Kembali ke Berand
            <a href="{{ route('peminjaman') }}" class="cta-btn cta-secondary">📋 Mulai Peminj
          </div>
        </div>
        <div class="beranda-hero-art">
          <div class="hero-book-icon">📖</div>
        </div>
      </div>

      <!-- Info Cards -->
      <div class="beranda-fitur">
        <div class="fitur-card">
          <div class="fitur-icon">🎯</div>
          <h3>Tujuan Aplikasi</h3>
          <p>Mempermudah petugas perpustakaan dalam mencatat, mengelola, dan memantau seluruh
        </div>
        <div class="fitur-card">
          <div class="fitur-icon">🛠️</div>
          <h3>Teknologi</h3>
          <p>Dibangun menggunakan <strong>Laravel</strong> (PHP Framework), Blade Template, CS
        </div>
        <div class="fitur-card">
          <div class="fitur-icon">🔔</div>
```

```
app > Http > Controllers > TentangController.php > ...
<?php

namespace App\Http\Controllers;

use Illuminate\Http\Request;

class TentangController extends Controller
{
    public function index()
    {
        return view('tentang');
    }
}
```

```
routes > web.php > ...
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\DashboardController;
use App\Http\Controllers\TentangController;
use App\Http\Controllers\BerandaController;
use App\Http\Controllers\PeminjamanController;
use App\Http\Controllers\DaftarController;

// Dashboard
Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');

// Tentang
Route::get('/tentang', [TentangController::class, 'index'])->name('tentang');
```

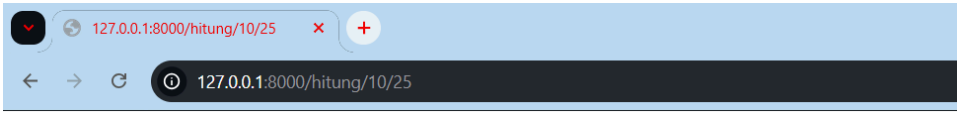
7. Jalankan php artisan route:list - amati semua route yang terdaftar

```
PS C:\xampp\htdocs\perpustakaan-pweb> php artisan route:list

  GET|HEAD  / ..................................................... beranda › BerandaController@index
  GET|HEAD  daftar ................................................ daftar › DaftarController@index
  GET|HEAD  dashboard ....................................... dashboard › DashboardController@index
  GET|HEAD  peminjaman ................................... peminjaman › PeminjamanController@index
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesystem/File…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/Filesyst…
  GET|HEAD  tentang ............................................ tentang › TentangController@index
  GET|HEAD  up vendor/laravel/framework/src/Illuminate/Foundation/Configuration/ApplicationBuilder…
```

Disini saya sudah menyesuaikan semua aktivitas mandiri sesuai dengan tampilan project yang saya buat yaitu Sistem Informasi Perpustakaan

8. Coba buat Route::get('/hitung/{a}/{b}', fn($a, $b) => $a + $b) dan akses /hitung/10/25

```
// Route dengan parameter — contoh Aktivitas Mandiri 2 no. 8
Route::get('/hitung/{a}/{b}', fn($a, $b) => $a + $b);
```



35

Penerapan ke Project Perpustakaan

File	Lokasi	Keterangan
web.php	routes/	Semua route perpustakaan
BerandaController.php	app/Http/Controllers/	Controller halaman beranda
PeminjamanController.php	app/Http/Controllers/	Controller form peminjaman
DaftarController.php	app/Http/Controllers/	Controller daftar buku
DashboardController.php	app/Http/Controllers/	Controller dashboard admin
TentangController.php	app/Http/Controllers/	Controller halaman tentang
layout.blade.php	resources/views/	Template utama (header, nav, footer)
beranda.blade.php	resources/views/	Halaman beranda
peminjaman.blade.php	resources/views/	Form peminjaman buku
daftar.blade.php	resources/views/	Daftar dan tabel peminjaman
dashboard.blade.php	resources/views/	Dashboard admin
tentang.blade.php	resources/views/	Halaman tentang
style.css	public/css/	CSS dari project HTML sebelumnya
script.js	public/js/	Javascript dari project HTML sebelumnya